COMPUTATIONAL DESIGN · DISASTER RESPONSE · OPEN DATA

# S.U.R.G.E.

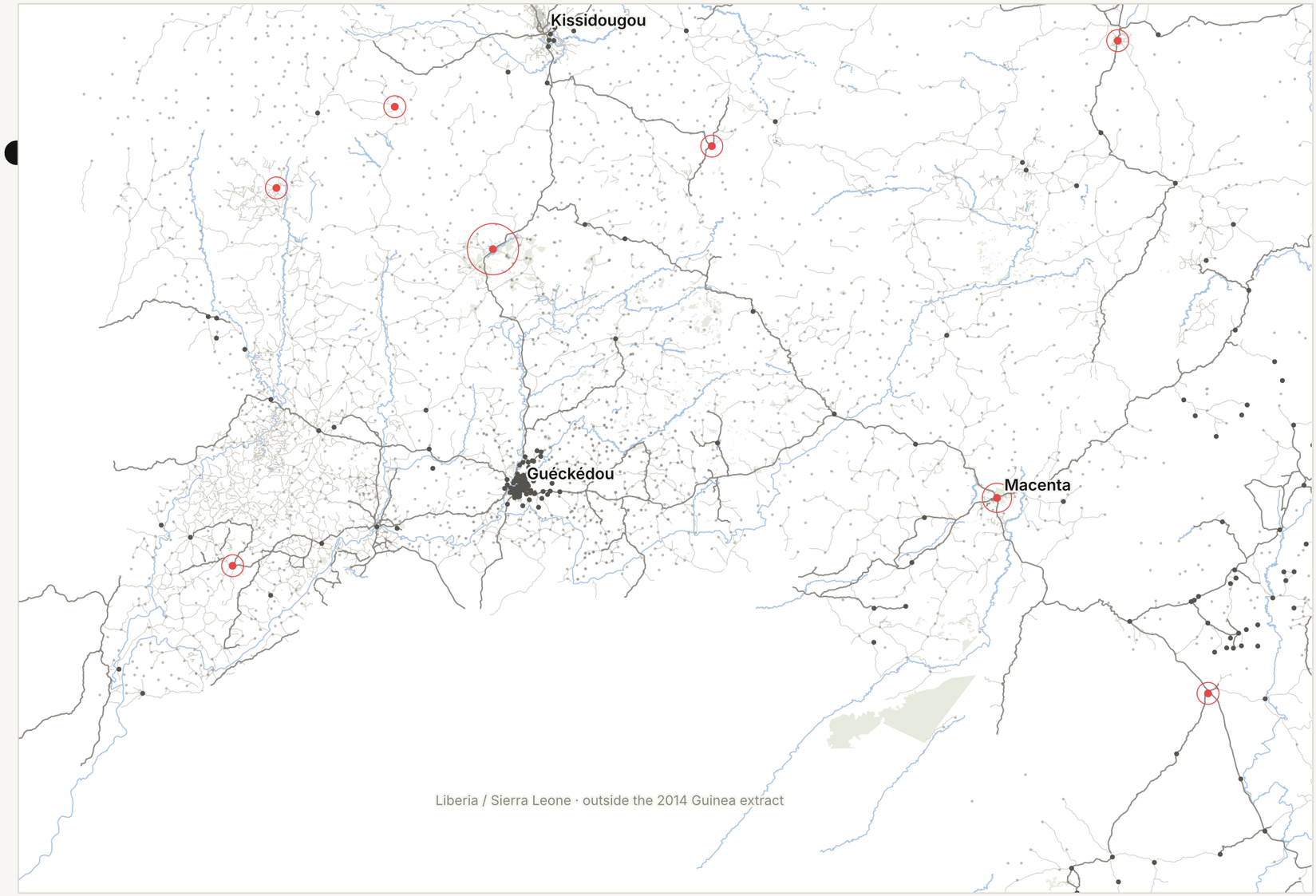
## Spatial Unit Response Generator for Epidemics

A disaster produces a flood of data; S.U.R.G.E. turns that data into treatment sites, ward sizes and ready-to-assemble wards, automatically.

Rebuilt in 2026 from the author's 2015 studio project, the Parametric Response System (P.R.S.).

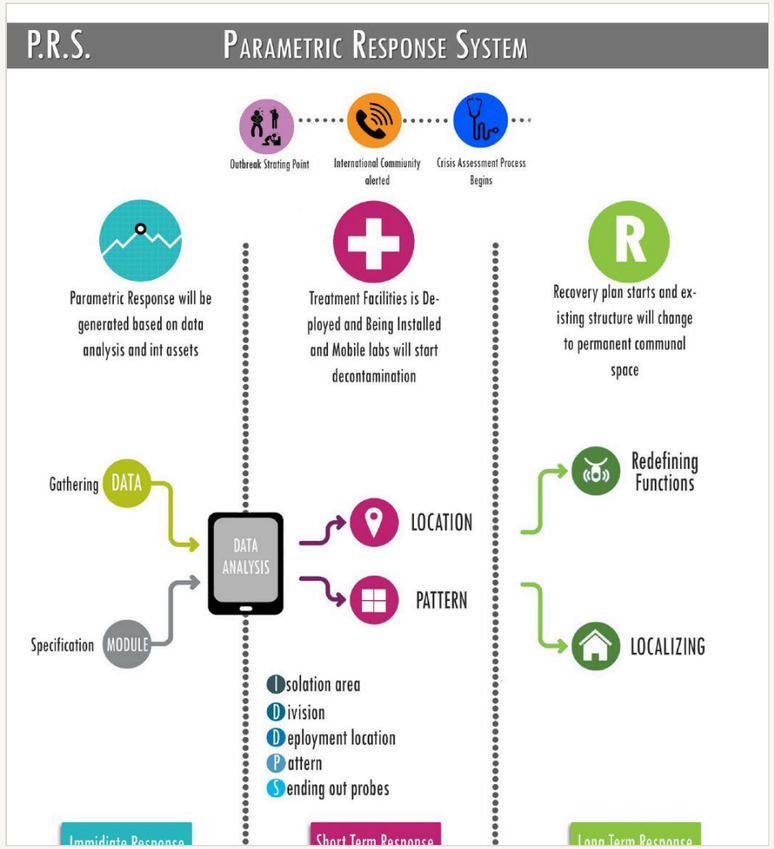
| | |
|---|---|
| STUDY AREA | Guéckédou · Macenta · Kissidougou, Forest Guinea, where the 2014 West Africa outbreak began |
| INPUTS | OpenStreetMap 2014: 2,435 settlements, 18,028 road segments, waterways, land use, forest |
| OUTPUTS | Outbreak forecast → catchments → treatment sites → ward plans → after-use |

Shahriar Bigdeli · Python + Grasshopper · origin: Design 5A ARCT5500, UWA Summer Studio 2015

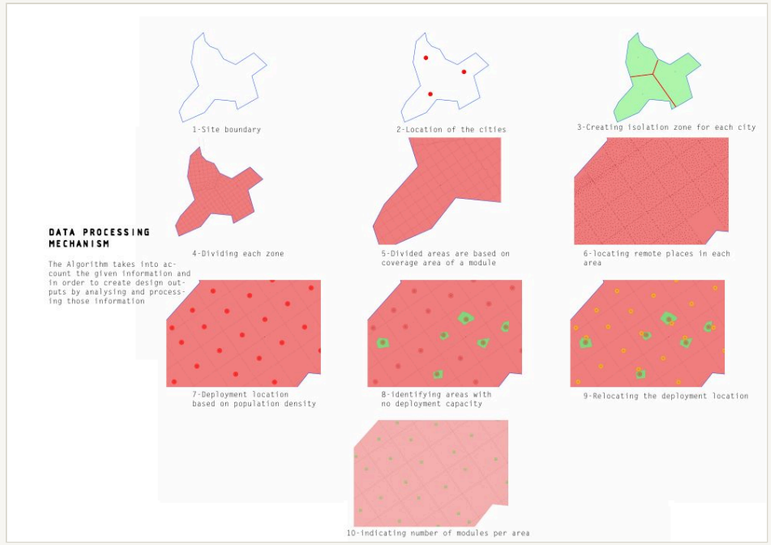



Prototype output on real 2014 OSM data: red rings are the 8 computed treatment sites, sized by number of 12-bed modules.

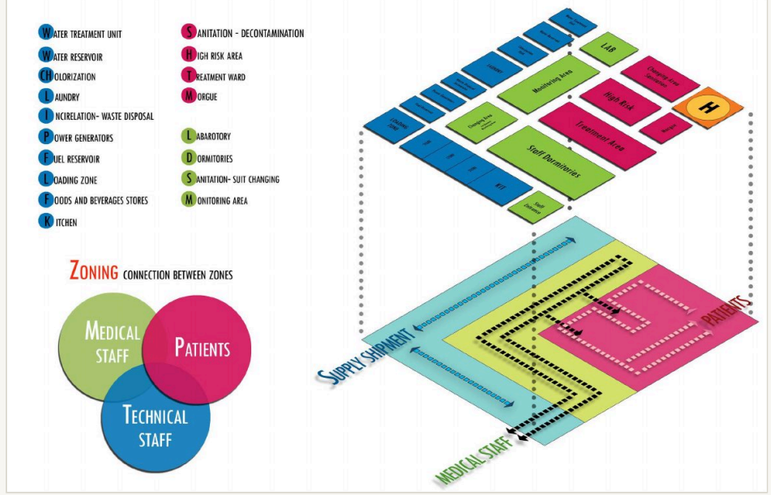
# Where this started




2015 · The P.R.S. diagram: data + module spec → analysis → location + pattern, then deploy, then convert.




2015 · The 10-step Grasshopper “data processing mechanism” (MeerkatGIS + Voronoi + grid).




2015 · Ebola treatment unit layout and zoning.

### WHAT HELD UP

- Framing the response as a **pipeline**: data goes in, a spatial plan comes out.
- Three phases: **immediate** (analyse), **short term** (deploy), **long term** (convert to community space).
- Grounded in real field references: MSF three-ward ETUs, the Monrovia Medical Unit, CDC Firestone.

### WHAT WAS THIN

- Epidemiology was two constants ($R_0$ = 2, 0.3%). Nothing changed over time.
- Isolation zones were **straight-line Voronoi** cells. They ignored roads.
- Deployment was a **regular grid** with manual relocation. Nothing was optimised.
- Ward plans weren't driven by predicted case load.

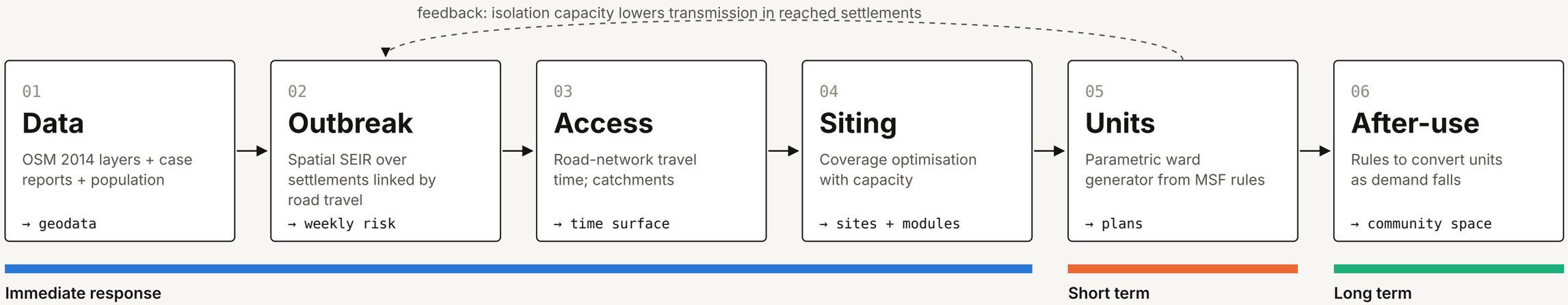
### THE REWORK

Keep the idea and the narrative. Replace each step with a model that can be tested.

> **YOU · T01** Re-export the 2015 boards at full resolution (the .ai files in /New folder) for this spread.

# From diagram to pipeline

Each 2015 step becomes a model with explicit inputs, assumptions and outputs. The phases are the same as the 2015 P.R.S. diagram.




## 2015 → 2026 MAPPING

| 2015 step | Rework |
|---|---|
| Contagious factor = 2 | Spatial SEIR model, run weekly |
| 3 · Voronoi isolation zones | Road travel-time catchments |
| 4–5 · Grid of 500 km² cells | Coverage optimisation |
| 7–9 · Density + manual relocation | Demand-weighted siting with constraints |
| 10 · Modules per area | Beds from forecast → 12-bed modules |

## STACK

Python · numpy · road graph + Dijkstra · SEIR (metapopulation) · greedy max-coverage → MILP · Rhino/Grasshopper for the unit · web map for the interface

**Locked:** Python for data → location → ward sizing; Grasshopper (linked with Hops) for the automatic ward design.

## REPRODUCIBILITY

One command rebuilds every figure from the raw shapefiles in `References/Data`. Each page notes which parameters drive it.

**YOU · T02** Put the pipeline in a public repo with a README, pinned environment and licence (OSM data is ODbL).

# What the system reads

The six OSM layers below (Sept–Dec 2014) came with the 2015 project, clipped to the same window as the 2015 MeerkatGIS crop. Step 1 added population, terrain, land cover, boundaries and case data (right; full atlas in `rework/data/atlas_step1.png`).



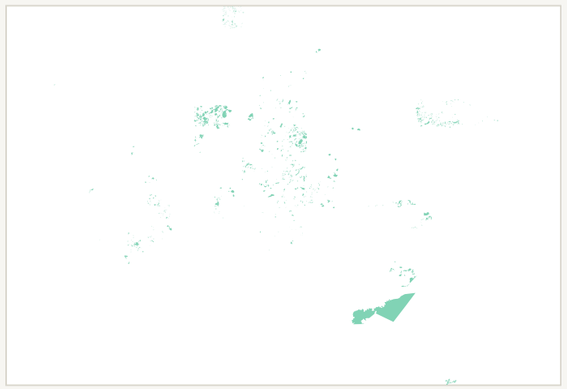
**Settlements** 2,435 points · city→hamlet



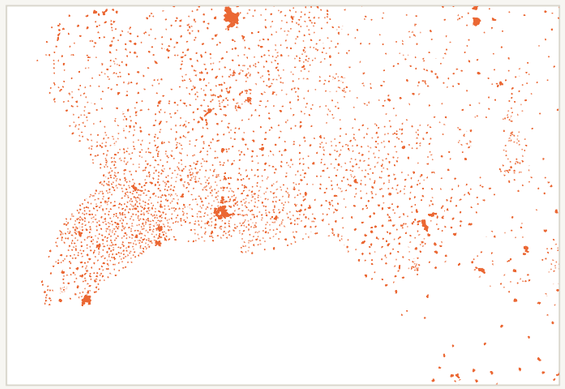
**Roads** 18,028 segments · 12 classes



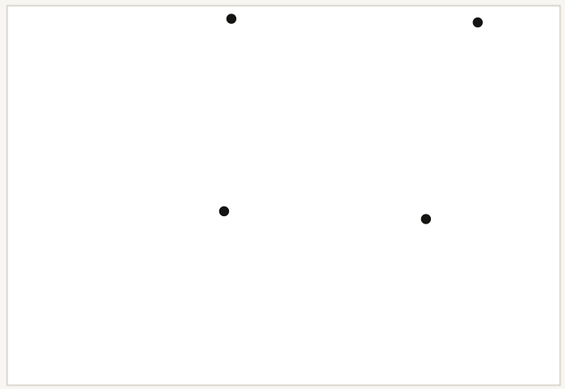
**Waterways** rivers, streams · crossings matter



**Forest / natural** exclusion layer for siting



**Residential land use** isolation buffer source



**Cell towers** 5 sites · reporting reach

## GAPS TO CLOSE

**Step 1 done:** HRSL population (90 m), Copernicus DEM + slope, ESA WorldCover land cover, geoBoundaries admin units, Guinea MoH sitreps by prefecture (Aug–Oct 2014) and WHO weekly cases. See `rework/data/SOURCES.md`.

**Added:** 172 MoH health facilities in the window and CHIRPS 2014 monthly rainfall (2,294 mm/yr, peak July).

**Facility data repaired:** 207 facilities in the window. 172 are from the MoH list, 13 matched by village name, the Guéckédou hospital is placed at its town centre and 21 are scenario placements. See `data/facility_repair.png`.

Cell coverage only has 5 tower points inside the window. Treat it as a proxy for where reporting is possible, not as true coverage.

# Where the virus goes, week by week

Metapopulation SEIR over 115 settlement clusters. Clusters are coupled by a gravity model on **road travel time**, so the virus moves the way people do. Seeded at Guéckédou, near Meliandou, where the index case was.



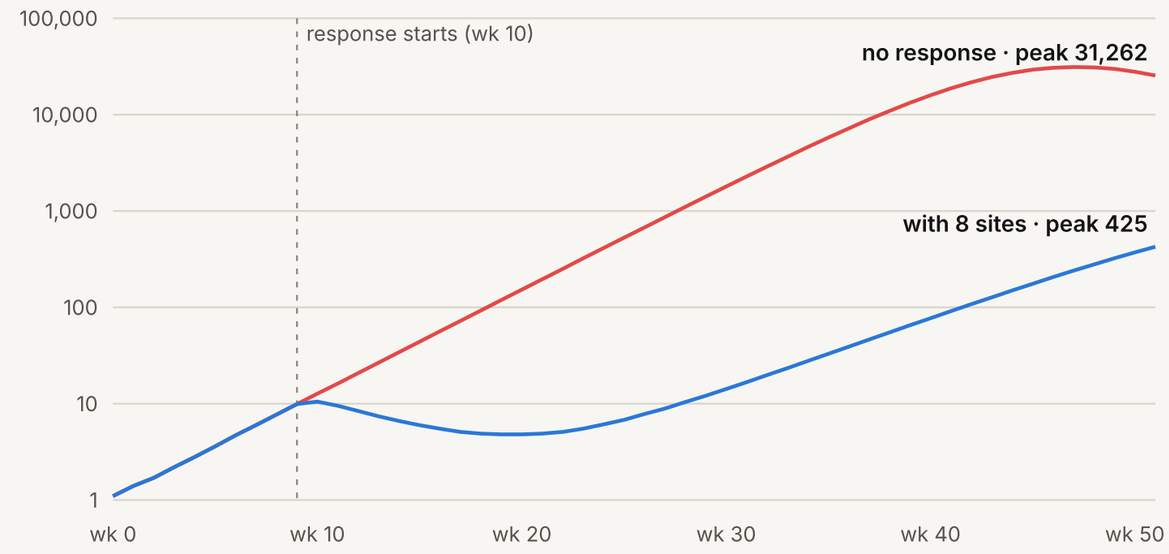
**Week 10** 16 infectious **Week 20** 198 infectious **Week 30** 2,343 infectious **Week 40** 22,163 infectious

infectious per cluster (log) circle size = population proxy

$$dS = -\beta \cdot S \cdot (M \cdot I/N) \cdot dE = \beta \cdot S \cdot (M \cdot I/N) - \sigma E \cdot dI = \sigma E - \gamma I$$

$$R_0 = 1.7 \text{ (lit. } 1.5\text{–}2.0) \cdot 1/\sigma = 9.4 \text{ d} \cdot 1/\gamma = 7.4 \text{ d} \cdot M = 0.92 \cdot I + 0.08 \cdot \text{gravity}(P, e^{-t/2h})$$

**BED DEMAND, REGION TOTAL**



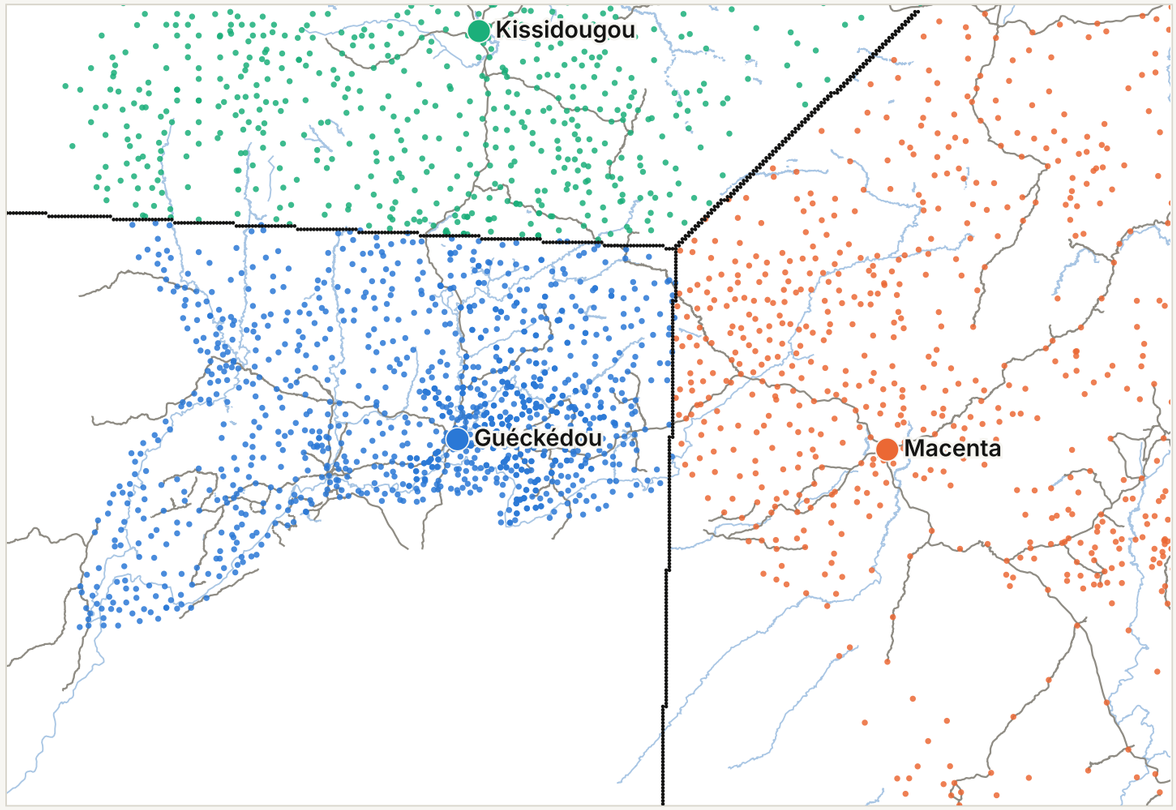

Assumes 60% of infectious people seek a bed. Isolation within reach of a site cuts transmission by 60%. **Uncalibrated**: this shows the mechanism, not a forecast. The real outbreak was far smaller and much more stochastic.

**Done (step 2):** calibrated to the MoH prefecture sitreps: $R_0$ 1.56, fit error 0.096 (log RMSE). See `data/s2/fig_1_calibration.png`.

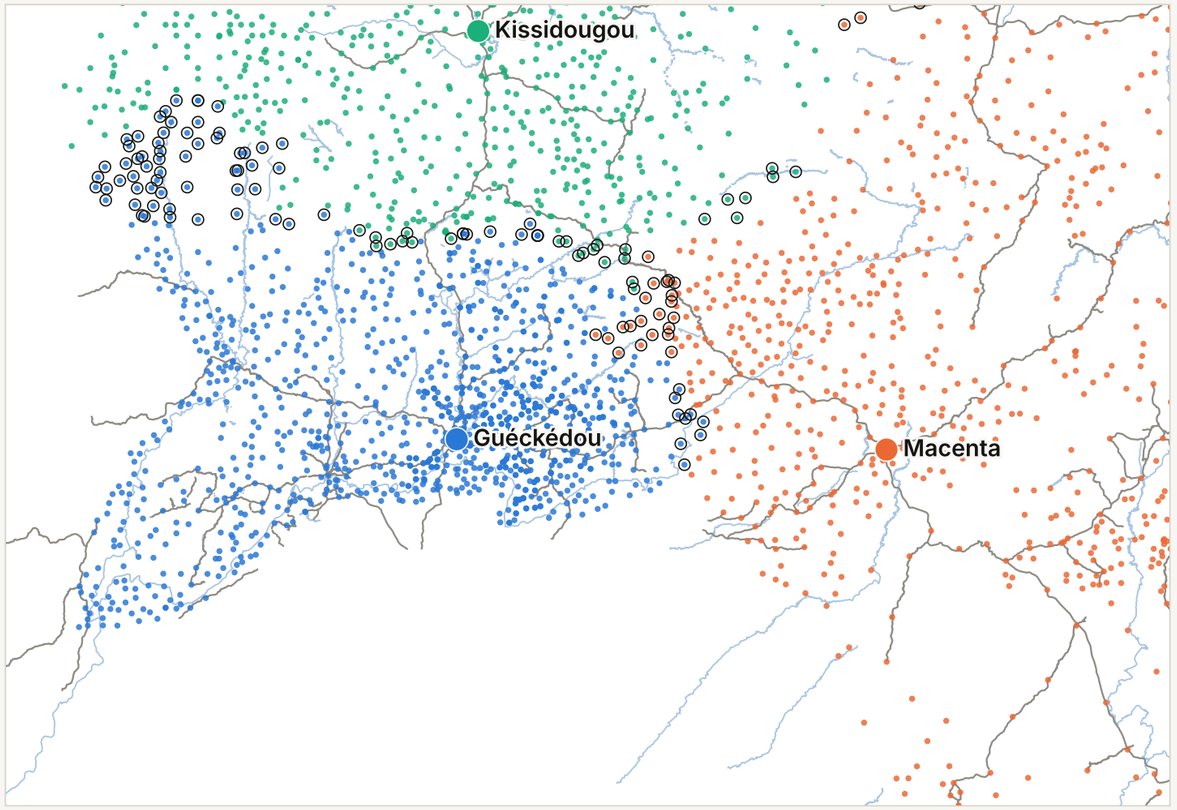
**YOU · T03** Make the model stochastic (e.g. τ-leaping, 200 runs) and show uncertainty bands instead of one line.

**YOU · T04** Add funeral transmission and health-care-worker infection as separate compartments. Both drove the 2014 outbreak.

# Distance is travel time, not straight lines




**2015 · straight-line Voronoi.** Each settlement goes to the nearest city as the crow flies.




**Rework · road travel time.** Each settlement goes to the city it can reach fastest by road. Black rings mark settlements that switch catchment.

**128**

of 2,435 settlements (5%) change catchment once roads are considered

Guéckédou Macenta Kissidougou

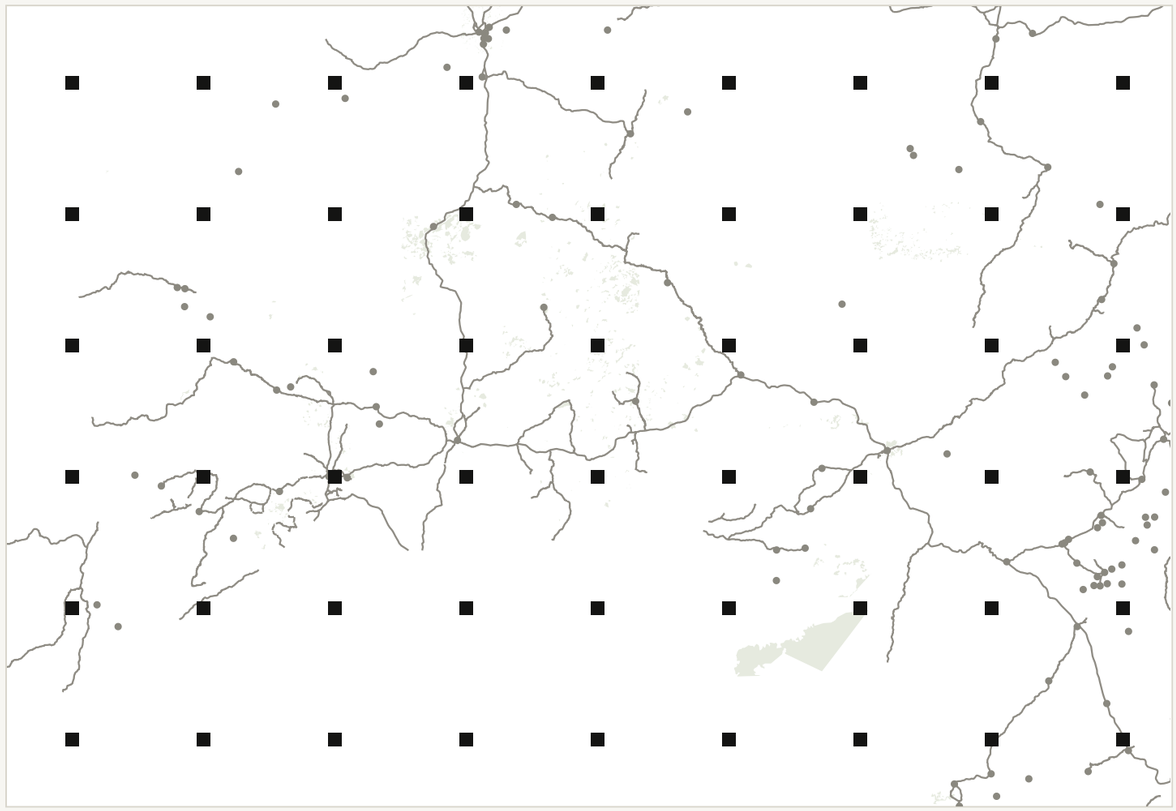
## HOW IT WORKS

18,028 OSM segments become a graph of 188k nodes. Speeds are set per road class (trunk 50 km/h → footpath 4 km/h). Multi-source Dijkstra from each city gives travel time and catchment for every settlement in one pass. Settlements off the network walk to the nearest node at 4 km/h.

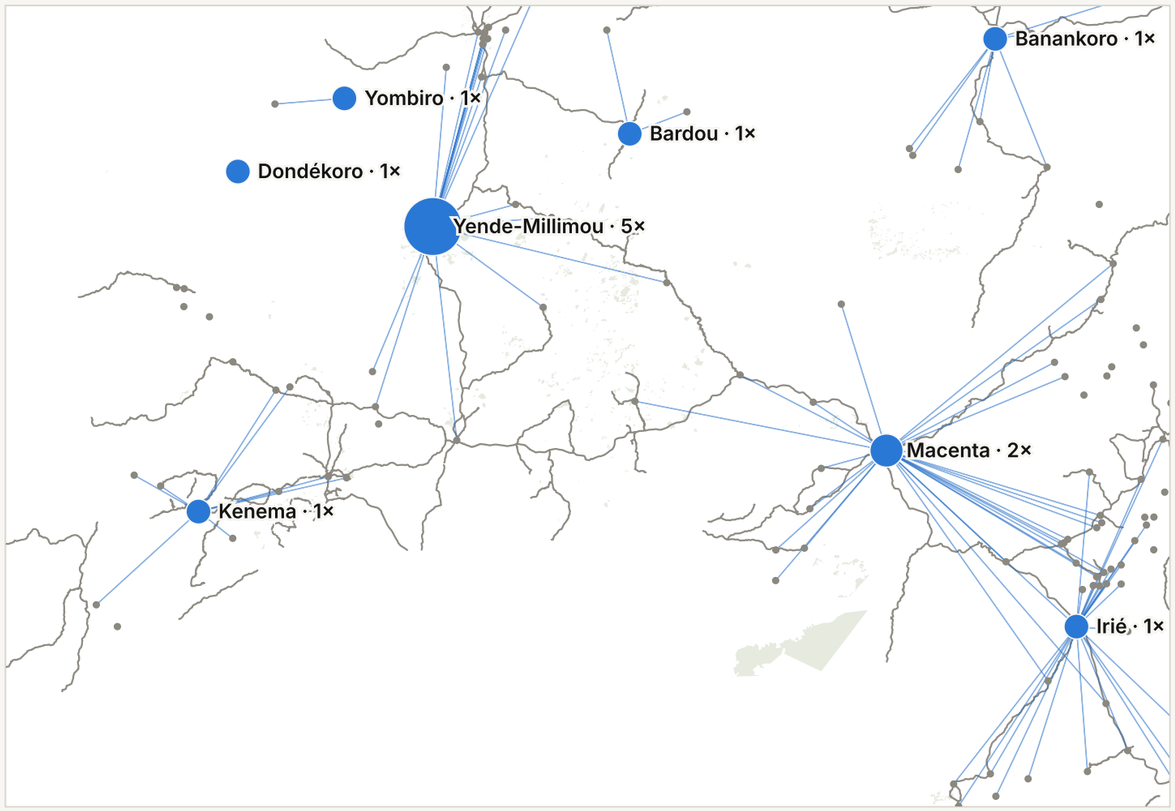
**YOU · T05** Check the road-speed table with someone who knows Forest Guinea. Add a rainy-season scenario where tracks and paths drop to about 50% speed.

**YOU · T06** Treat rivers without a bridge as barriers: intersect the waterways layer with roads and add ferry penalties.

# Fewer units, placed where they're needed



**2015 · one unit per 500 km² cell.** 54 units on a regular grid, no matter where people live.




**Rework · demand-weighted coverage.** Greedy maximum coverage: each site serves clusters within 1.5 h by road. Labels give the number of 12-bed modules, sized from that catchment's forecast peak.

**8**

sites cover 96% of peak demand within 1.5 h

**54**

grid units to reach 95%, the 2015 method

**0.9 h**

mean demand-weighted travel time (0.8 h for the grid, but with 7× the units)

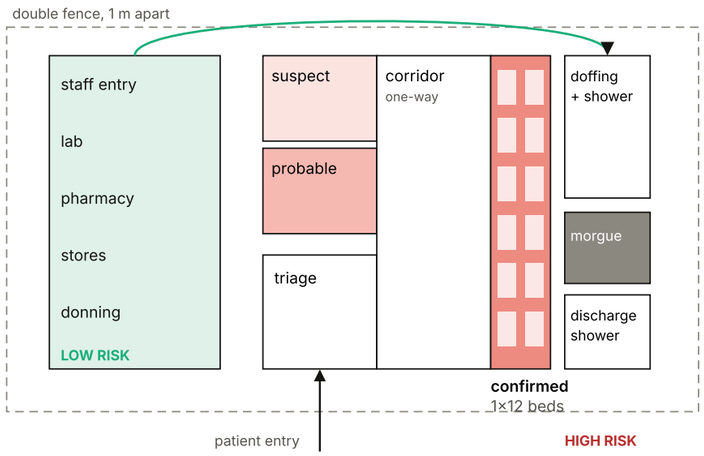
**Done:** capacitated maximal-covering MILP (HiGHS). 4 sites cover 97% of peak bed demand within 1.5 h; the 2015 grid needed 33 units.

**Done:** 8 MUST filters (68,885 → 936 candidates) and 9 weighted SHOULD scores, all in `s2_params.py`.

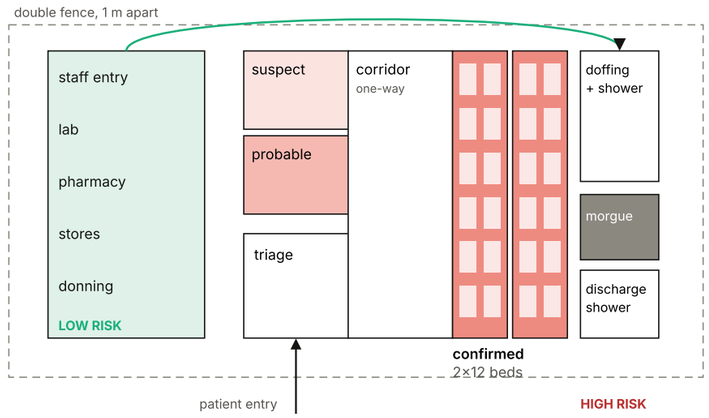
**Done:** sweep over 2–10 sites × 0.75–3 h, plus a facility-repair sensitivity. See `data/s2/fig_4_tradeoffs_feedback.png`.

# Plans generated from demand

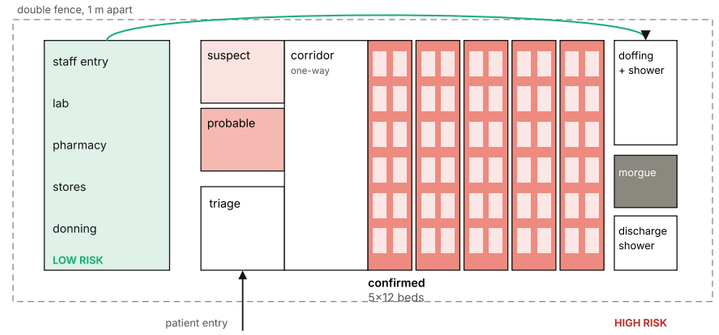
The generator turns a site's bed demand into a plan. The ward sequence and circulation rules come straight from the MSF three-ward model in `References/An Ebola treatment centre2.docx`. Only the confirmed ward grows; the rest stays fixed.




**1 module · 12 beds** · e.g. Dondékoro




**2 modules · 24 beds** · e.g. Macenta




**5 modules · 60 beds** · e.g. Yende-Millimou

### ENCODED RULES

- Sequence: triage → suspect → probable → confirmed → morgue / discharge
- All paths fenced and one-way; staff loop enters green, exits through doffing
- Foot basin + shower at every red→green crossing
- Double fence at 1 m; separate patient and staff gates
- Module = 12 beds (2015 brief); lab within the 500 km² coverage of the mobile lab

### PARAMETERS

| | |
|---|---|
| `beds_needed` | from stage 04, per site |
| `module_beds` | 12 |
| `staff_ratio` | **YOU · T07** check the draft ratios in pipeline/s3_params.py against MSF/WHO guidance |
| `plot_bounds` | from siting constraints |

**YOU · T08** Open grasshopper/surge_draw_layouts.py and the Hops server in Rhino 8, check the geometry, then swap the plain extrusions for the modelled origami tent.

**YOU · T09** Bring in the 2015 origami tent and negative-pressure module as the 3D kit of parts. Re-model it cleanly in Rhino.

**YOU · T10** Render one hero axonometric of the largest site (deployed camp, as on the 2015 'Deployed unit' board).

# The long-term response

The long-term phase from 2015 stays in. Once a site's forecast demand stays at zero, its modules are reassigned. The fixed parts (slab, water treatment, power) become the base for a community centre run for the families affected.

Weeks 0–4
### Deploy
Modules + mobile lab

Weeks 4–30
### Operate
Wards grow and shrink with the forecast

Weeks 30–42
### Decommission
Decontaminate · 42 days, no new cases

Year 1+
### Convert
Learning centre · workshop · garden

**LEARNING CENTRE**



2015 · Learning centre sketch

**WORKSHOP**



2015 · Workshop sketch

#### CONVERSION RULE (DRAFT)

```
if demand(site, t) == 0 for 42 days:
  wards → classrooms / workshops
  green zone → admin + counselling
  water unit → kept; garden on decontaminated slab
```

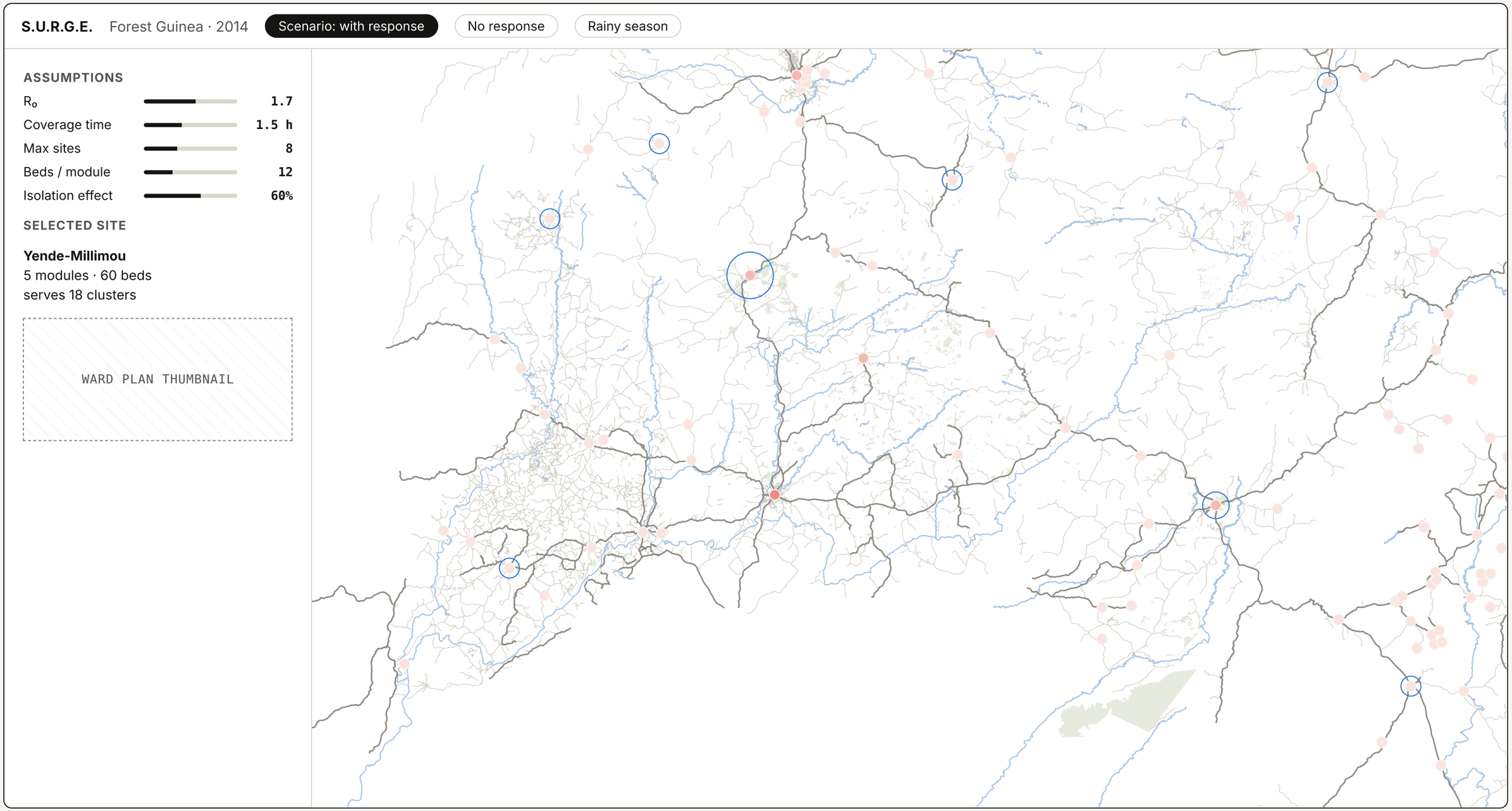
**YOU · T11** Write the conversion as rules, using the 2015 workshop table (learning centre 1:20, counselling 3×/wk, gardening, cooking, carpentry) as the brief for each programme.

**YOU · T12** Check the 42-day end-of-outbreak rule and the decontamination steps against WHO guidance, and cite them.

**YOU · T13** Redraw the two sketches in the same line style as the new plans, or keep them as-is and label them as 2015 work.

# A tool, not a board

The working output is an interactive map, not a static board. A reviewer can scrub through time, change the assumptions and watch the sites and wards update.




**YOU · T14** Build the web map (MapLibre or deck.gl). Precompute scenarios as JSON so it runs as a static page.

**YOU · T15** Record a 60–90 s screen capture of the slider and scenario switch for the portfolio and LinkedIn.

**YOU · T16** Decide whether the live tool is public. If so, add a clear 'research prototype, not for operational use' disclaimer.

# Testing the model against 2014

## MODEL VS. WHAT ACTUALLY HAPPENED

| Computed site | Modules | Prefecture | CTE there in 2014 sitreps? |
|---|---|---|---|
| Yende-Millimou | 5×12 | — | — |
| Macenta | 2×12 | — | — |
| Kenema | 1×12 | — | — |
| Banankoro | 1×12 | — | — |
| Irié | 1×12 | — | — |
| Bardou | 1×12 | — | — |
| Yombiro | 1×12 | — | — |
| Dondékoro | 1×12 | — | — |

**YOU · T17** Compare computed sites with the prefectures where CTEs operated in the 2014 sitreps (Guéckédou, Macenta). Discuss any site with no counterpart.

CHART: MODELLED VS OBSERVED WEEKLY CASES (AFTER CALIBRATION)

## LIMITATIONS, STATED PLAINLY

- OSM in 2014 was incomplete in rural Guinea. Road gaps bias travel times upward.
- Population is a proxy until WorldPop is added.
- The SEIR model is deterministic and uncalibrated. It overstates spread by orders of magnitude.
- Greedy siting is not optimal. It is only a baseline for the MILP.
- Community trust, burial practices and security shaped the real response, and none of them are in the model.

## REFLECTION: 2015 → 2026

**YOU · T18** Write 150–200 words in your own voice: what you'd now do differently, what computational design means to you, and how it shows in this rework.

**YOU · T19** Get one outside reviewer (epidemiologist or humanitarian logistician) to read pages 05–07 and quote their feedback.

# What's left for you

Every yellow **YOU** marker in this mockup, collected here. Everything else on these pages, including the maps, catchments, outbreak curves and siting numbers, was **computed from the 2014 data already in this repo** and is a working prototype you can build on.

### MUST DO

- ☐ **T08** Test the generator in Rhino 8 + model the tent — p.08
- ☐ **T14** Build interactive web map — p.10
- ☐ **T17** Compare sites with 2014 CTE prefectures — p.11
- ☐ **T18** Write personal reflection (150–200 words) — p.11

### SHOULD DO

- ☐ **T01** Hi-res exports of 2015 boards — p.02
- ☐ **T02** Public repo + README + ODbL attribution — p.03
- ☐ **T03** Stochastic runs + uncertainty bands — p.05
- ☐ **T05** Validate road speeds + rainy-season scenario — p.06
- ☐ **T07** Verify step 3 ratios (staff, water, areas) — p.08
- ☐ **T09** Re-model origami tent + HVAC module in Rhino — p.08
- ☐ **T10** Hero axonometric of largest site — p.08
- ☐ **T11** Formalise conversion rules from 2015 programme table — p.09
- ☐ **T15** Screen-capture video of the tool — p.10
- ☐ **T16** Public tool? + disclaimer — p.10
- ☐ **T19** External expert review + quote — p.11

### NICE TO HAVE

- ☐ **T04** Funeral + HCW compartments — p.05
- ☐ **T06** River crossings / ferry penalties — p.06
- ☐ **T12** Cite WHO end-of-outbreak + decon guidance — p.09
- ☐ **T13** Decide: redraw or keep 2015 sketches — p.09

### SUGGESTED ORDER

1 data gaps → 2 calibrate → 3 MILP siting → 4 Grasshopper units → 5 web map → 6 boards + reflection

### LEGEND

**YOU · Txx** task = needs you

PLACEHOLDER

### REGENERATE

```
cd rework/mockup/build
python3 compute.py && python3 render.py
node pdf.mjs  # needs playwright
```